Requirements (Which of the Goals does this Complete?):

Collect FUEL in AUTON from GROUND
Collect FUEL in Teleop from Ground

Concepts:

- "The grabby Arm"
  - Pros: it looks fun
  - Cons:one at a time
- Single Roller Bumper Gap Intake Ex. Citrus Circuits Season Recap 2017
  - Pros:
    - SIMPLE
  - Cons:
    - Max 2.9 Balls Wide, 17.5in (Assuming Square Robot)
    - Gets Hit Easily
- Multiroller Bumper Gap Intake Ex. Team 118 Robonauts 2017
  - Pros:
  - Cons:
    - Max 2.9 Balls Wide, 17.5in (Assuming Square Robot)
    - Gets Hit Easily
- Slapdown Multiroller (Directly Under the Storage/Hopper) Ex 254. 2017
  - Pros: WIDE intake, 25-27 in (4.5ish balls wide)
  - Can be made to feed balls straight up (into an extended Hopper)
    - Can be linked to an extending hopper system
  - Cons:
    - Complex Compared to Other Options
    - Weight
- "The Wall" Slapdown?- Ex. 1690 2020
  - Pros: WIDE intake, 25-27 in (4.5ish balls wide)
  - Cons:
- "The Wall" 4 bar - Ex. 1678 2020
  - Pros: WIDE intake, 25-27 in (4.5ish balls wide)
  - Cons: "I'm a delicate Flower"

Prototyping:

Compression?
Intake Wheel Material?

Examples:
2024 604
Something Beefy

HOOPER EXTENSION

- **No extension**

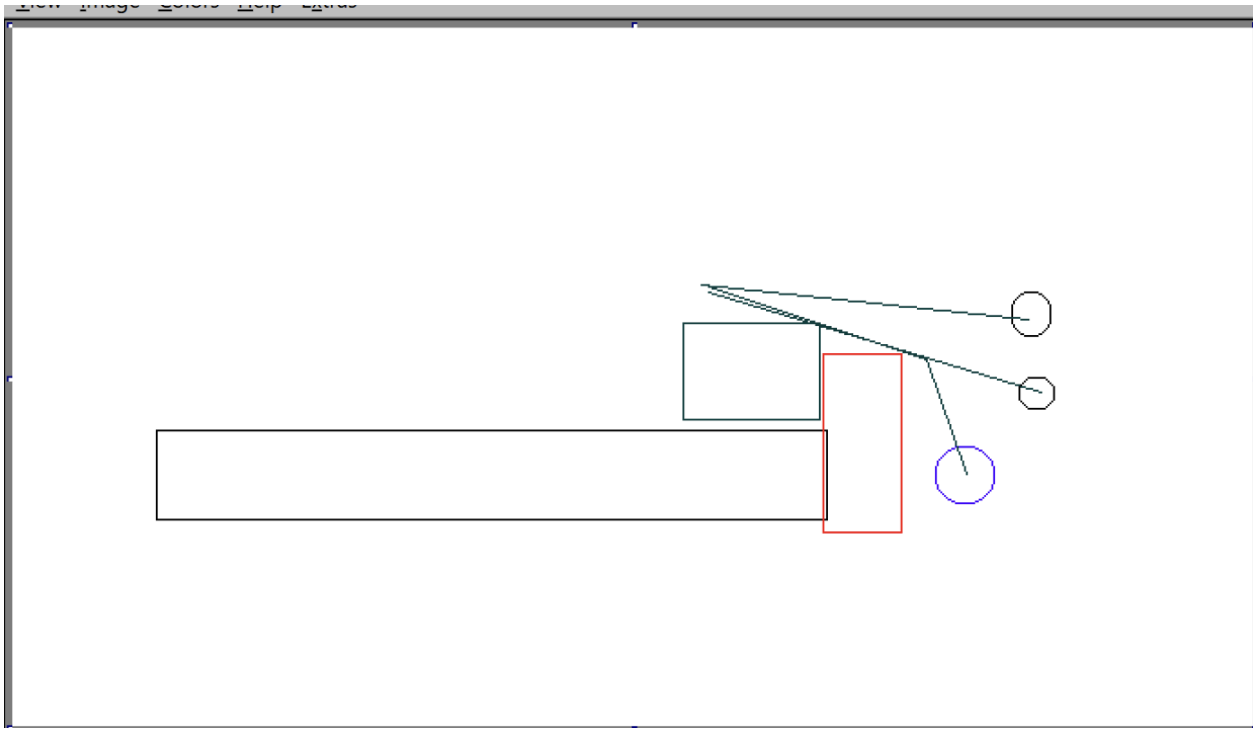
Passive Extension

Indipendent Extension

Linked Intake + Hopper Extension

Hooper = Intake

- **Rack and pinion**



Requirements (Which of the Goals does this Complete?):

- Ferrying from neutral zone
- Scoring into hub
- Holding pieces
- Scoring as much fuel as possible as quick as possible (shorten cycle times)

Concepts:

- Fuel hopper; Similar to 1678 and 254 2017, open top that can also take from rollers on the base
    - **Pros:**
        - Lots of storage, could store a good bit of fuel, (Likely upwards of 40-50 fuel, (thanks Adi RI3D concept holding around 50 fuel))
        - Decently simple, like would rank this at a middle/medium difficulty
        - Works really well with outpost/human player station intaking due to open top, just pour as many as possible in and start shooting when at hub (and its active)
        - Fastest net cycle time (longer to fill, more to score)
    - **Cons:**
        - Takes up a lot of room on robot
        - Possible jams (more likely than other options)
        - A lot of belts/rollers to function
        - If it expands past the frame perimeter its going to get hit
        - Could cause issues going under trenches, have to keep it shorter or find a way to fold
- Spindexer; 1690 2020
    - **Pros:**
        - Less chance of jams (slots for fuel)
        - Efficient
    - **Cons:**
        - Limited fuel storage max 5 fuel
        - Takes up decent bit of space
        - Complex
        - Requires additional pathing/almost a second mini indexer to get to shooter (tower)
- “S- curve” Indexer; 2910 2021
    - **Pros:**
        - Takes up smaller space
        - Unlikely to jam, fuel follows a rigid path
        - Pretty simple
    - **Cons:**
        - Less fuel storage, 3 fuel max (maybe 4 but it would be difficult and the curve would be have to be pretty harsh (and tall))
- Multi roller indexer (Vertical and horizontal); 4414 2021

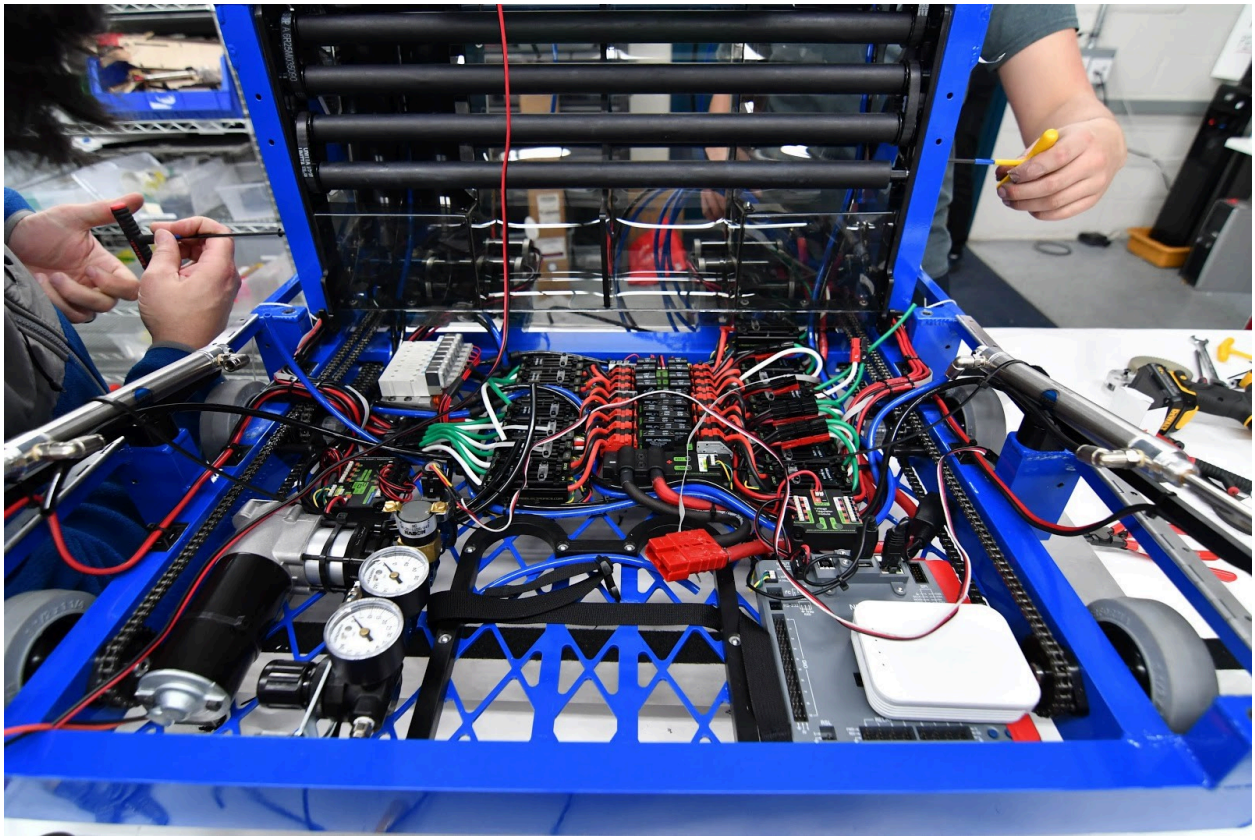
    - **Pros:**
        - Significantly less space taken up on robot
        - Really simple
    - **Cons:**
        - Little to no storage for fuel, max 1-2 fuel (likely 1)
        - Medium jam possibility (2nd most likely to jam)
        - Slow cycle times (due to limited fuel)

- Combo spindexer and hopper; 1986 2017 1986 Team Titanium 2017 Robot Reveal
    - **Pros:**
        - Lots of storage space (roughly equal to just the spindexer idea, maybe -1 or 2 fuel)
        - Not more complicated than the ideas individually, just basically a spindexer with hopper walls
        - Works with both ground intake and outpost intake from top
        - Cuts the amount of rollers/belts needed for just the hopper (replaces that with the spindexer portion)
        - Fast net cycle time (long reload but more ammo)
        - (seems) very efficient
    - **Cons:**
        - Still takes up a lot of room on robot
        - Still needs booster/ mini indexer to go to the shooter
        - Jams would be questionable???? (Taking most likely to jam and least likely to jam and putting them together so im unsure)
        - Expansion would still be a concern

Examples:

Ideas for morph chart:

- **Open ended hopper (best option)**
- **Spindexer**
- **Spindexer hopper combo**
- **“C/S-Curve” powered control path**
- **Power tower (L shaped path)**
- **Dual lane open hopper (just add a divider)**



- **Electrical possibility (254 2017, consider how hopper walls will affect the access to electronics if mounted to drivetrain)**
    - **Requirements**
        - **Make sure electrical has EASY access**
        - **Some sort of protection for the electronics, no way fuel can enter**
        - **Optimizes space**
        - **No way of short circuits**
        - **Indexer cannot fall onto electronics**
        - **Easily mountable to drivetrain**
        - **Not too difficult to play tetris with**
        - **Compatibility with intake**
        - **Easy to replace parts on both ends (electrical and mechanical)**
        - **Something to hold the hopper back (so it doesn't fall)**
        - **Gears do not cause wires to be severed or damaged**
        - **Tug test proof**
        - **Motor space**
        - **Wires not squished**
        - **Proper wire length**
        - **No interference with RSL**

- **Options**
  - **254 2017**
  - **Detachable bellypan (bellypan sandwich)**
  -

Requirements (Which of the Goals does this Complete?):

Scoring in HUB when active
Scoring in AUTON
Scoring in TELEOP when HUB is Active
Aligning to shoot into the HUB
Align to ferry from NEUTRAL ZONE
Ferrying from NEUTRAL ZONE

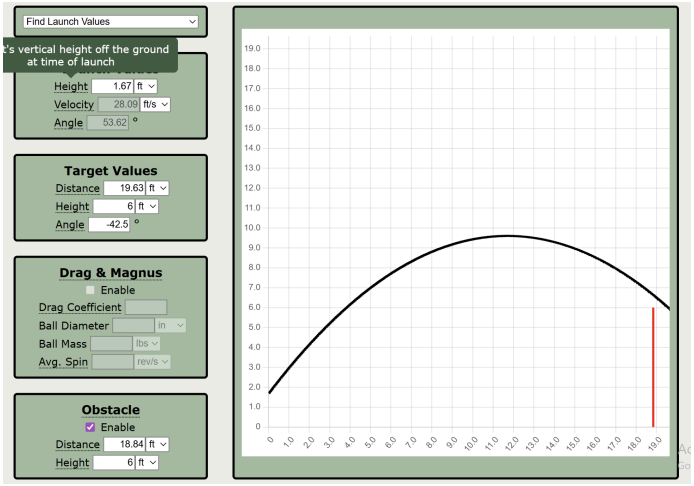
Shooting Numbers

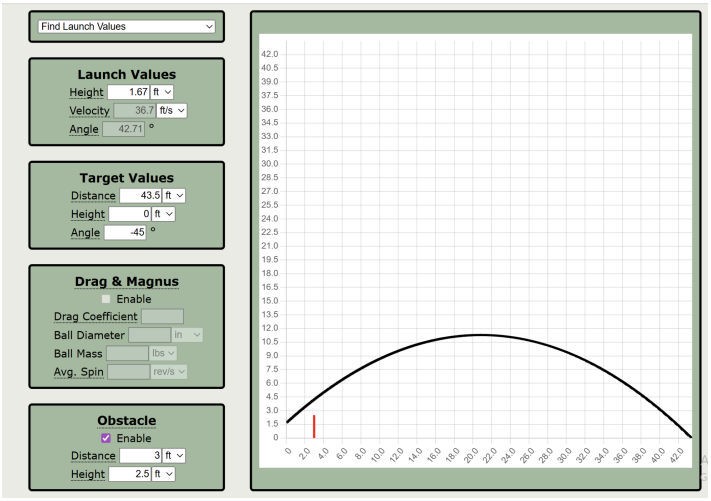
Targets:

- Height: 72in
- Angle : >33.690068deg(From horizontal)
- Corner Shot Into Hub Distance (along the ground): 238.758576895in
- Right Up Against the Hub Cenetered : 23.4999995in
- Far Neutral Ferry : 519.689023in (Fartherst point in neutral to Fartherst Point in Alliance Zone

Shot Type	Horisontal Distance	Target Height	Angle	Exit Velocity
Corner Shot Into Hub	238.758576895in	72in	>35degrees	28.09ft/s
Right Up Against the Hub Cenetered	23.4999995	72in	>35	20.17ft/s
Far Ferry	519.689023in	0	Any (clears a 30in obstacle up against the bumpers, defending robot)	36.7ft/s

w





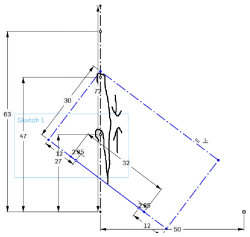
Concepts:
Dual Shooter - 1678 2017
Triple Shooter - Never done before, 148 2017 prototype
Examples:

Ideas:

Requirements (Which of the Goals does this Complete?):

L1 Climb
Climb besides somebody
Climb above somebody

Concepts:

- Vertical 2022 style climb 1678 (but 1 arm)
    - Pros
        - Existing Designs to base off of
        - Passive Grab
        - Pivotable (needed for multistage)
    - Cons
        - Pivoting needed (adds complexity potentially)
        - Needs to pop out at end of match or even beginning (l1 auton)
        -
- “Grab and yank”(???) climb Ex. 254 2013
    - 
    - **Pros**
        - Fast
        - Already done
        - Can be stowed
    - Cons
        - Needs room on robot
        - Lots of rung real estate
        - Added complexity two arms and two pivots
        - Robot would get in its own way(fatal)
- Simple L1 climb Ex. 610 2013
    - Pros
        - low complexity
        - Easy to implement
        - Less bot real estate
        - Fast
    - Cons
        - Only l1
        - Rung realestate
- Climb centered on UPRIGHT
    - Pros

        - No swing
        - Consistent
        - Compatible with previous ideas
        - Low run real estate
    - Cons
        - Pivot complexity
        - Precise alignment
        - No reference (maybe i just dk ball)
        - Robot has to go under the bar // no it doesnt
- “Fork climber”
    - Pros
        - easy
        - Fast
        - Low rung real estate
    - Cons
        - L1 only
        - Alignment

Examples: https://www.youtube.com/watch?v=eYPNndagILM

https://www.youtube.com/shorts/MVogy7rAPpU

Ideas:

Requirements (Which of the Goals does this Complete?):

Drive in Teleop
Drive In Auton
Drive over bump

*Fit all subsystems on it

Concepts:

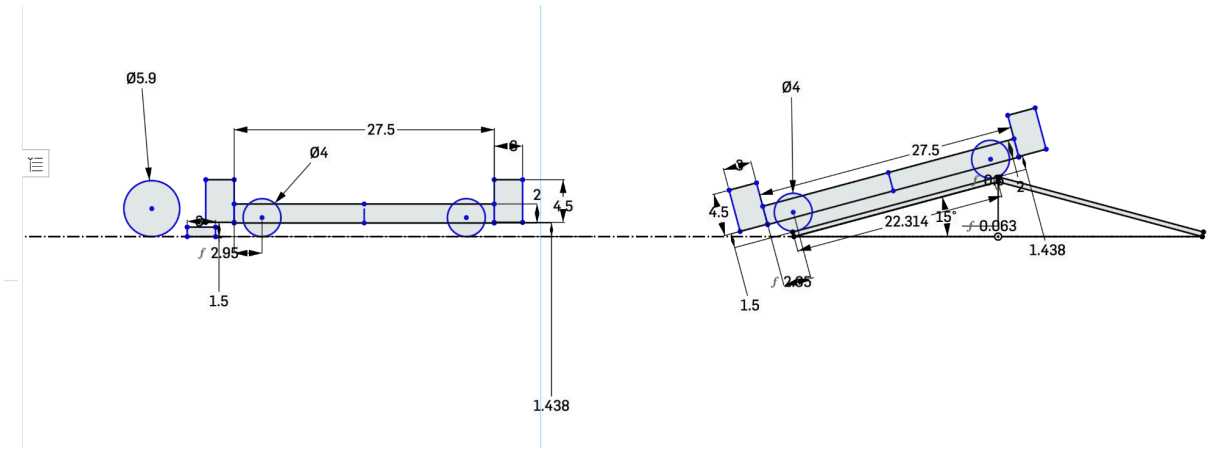
Examples:

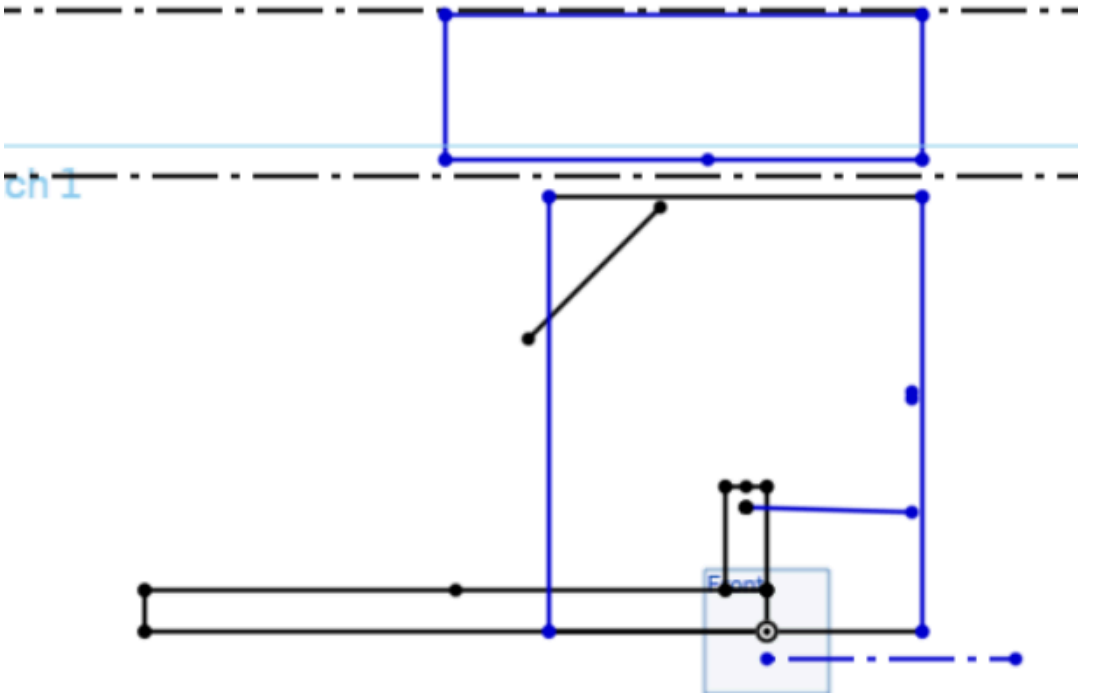
Ideas:

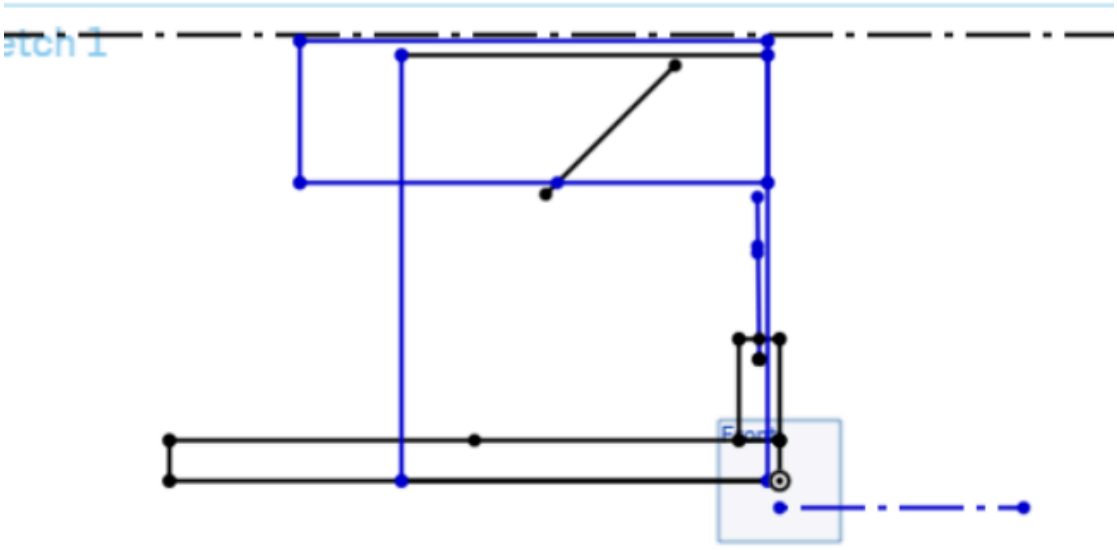
Bumper clearance sketch :

- 1.5” bumper to ground clearance
- 1.438” BOB (bottom of bot) to ground clearance
- 2.95” center of swerve wheel to edge of chassis
- 3” wide bumpers
- 4.5” tall bumpers
- 27.5” L x W robot




Ø5.9
27.5
Ø4
8
2
4.5
2.95
1.5
1.438
Ø4
27.5
8
2
4.5
22.314
15°
0.063
1.438
2.95
1.5


ch 1
Front


etch 1
Front